RF Link Budget Analysis

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Reference: “CubeSat Link Budget: Elements, Calculations, and Examples”
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MATLAB based interactive tool to analyse satellite communication links across modulation schemes and frequency bands

Key formulas for the Link budget calculation

The primary analysis is based on the **Friis free-space path loss** equation.

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2$$

Receiver noise is calculated using the **noise figure** definition, along with the **thermal noise floor** (−174 dBm/Hz). This is multiplied by the **bandwidth** to get the noise added by the receiver

$$NF = \frac{SNR_{in}}{SNR_{out}}$$

$$N_{dBm} = -174 + 10\log_{10}(B) + NF_{dB}$$

The received signal power is divided by the noise to get the SNR. The **energy per bit to noise density ratio** (E_b/N_0) is implicitly dependent on the modulation scheme used and explicitly on the datarate.

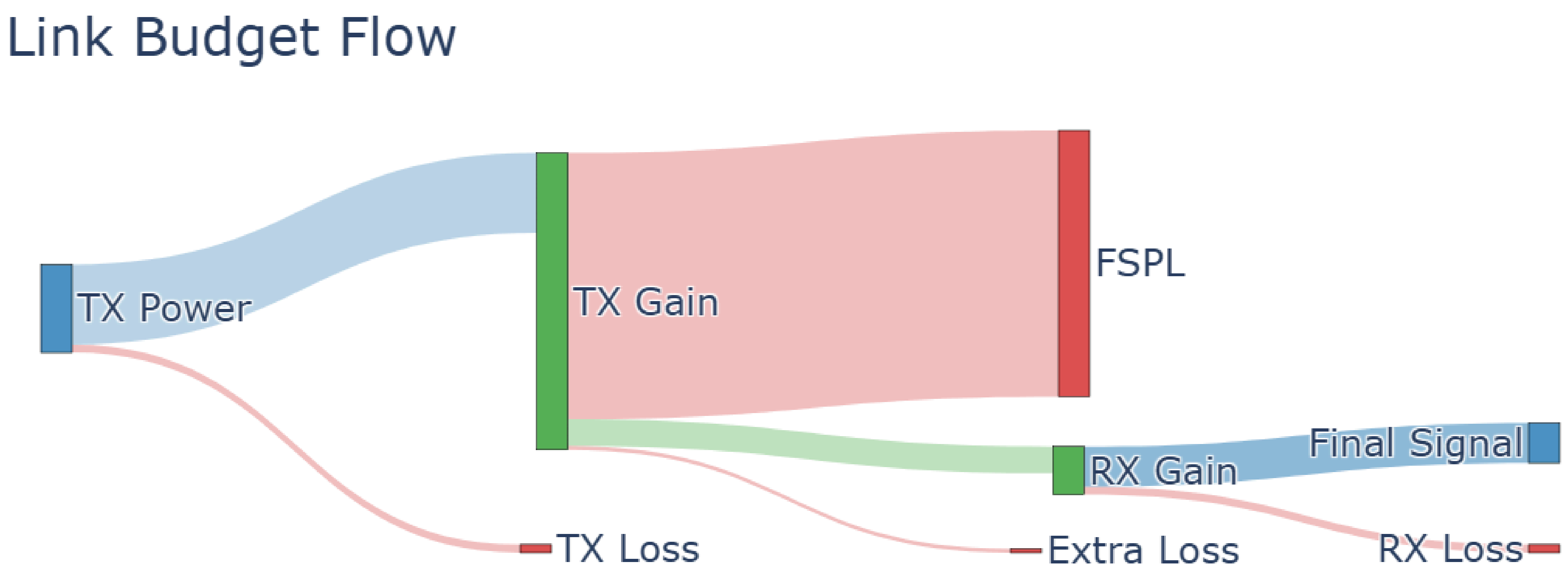
$$E_b/N_0 = P_r - (N_0 + 10\log_{10}(R_b))$$

Losses in the system

The following factors have been taken into account in the system model:

- Transmit Power
- Transmitter Gain
- Lumped transmitter loss (3 dB)
- Receiver Gain
- Lumped receiver loss (3 dB)
- Pointing loss $L_p(\mathrm{dB}) = -12 \times \left(\frac{\theta_e}{\theta_{\mathrm{BW}}}\right)^2$
- Atmospheric losses (1.5 dB)
- Noise figure of the Receiver
- Implementation loss (3 dB margin over min. required SNR)
- Free space path loss:

$$FSPL = G_t G_r \left(\frac{\lambda}{4\pi d}\right)^2$$



$$P_{rx} = P_{tx} + G_{tx} + G_{rx} \\ -(L_{tx} + L_{rx} + L_p + \mathrm{FSPL} + L_{extra})$$

Sensitivity to changes

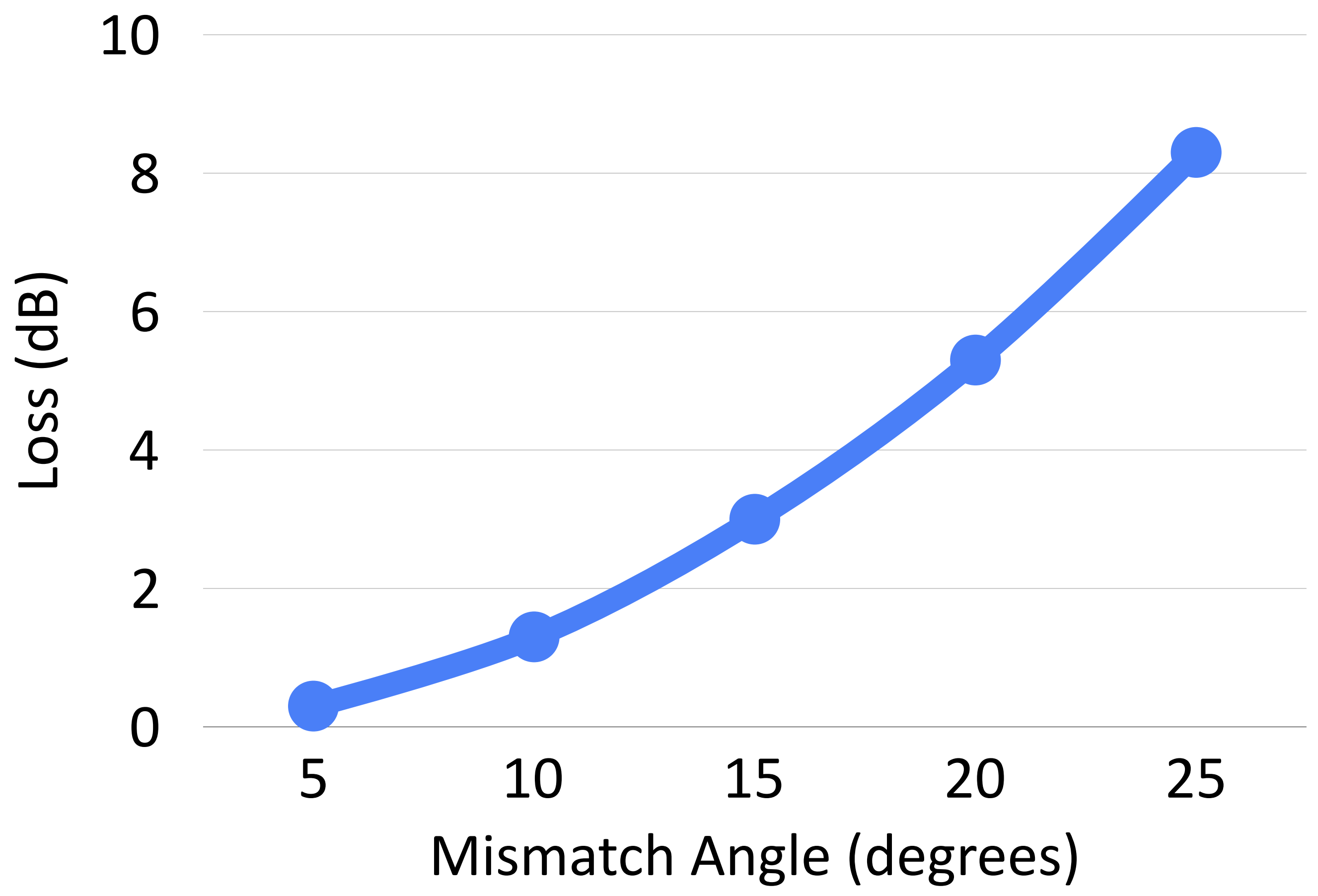
The table shows the effect of a change of 3 dB in each of the parameters on the final received power.

$$\Delta P_{rx} = \Delta P_{tx} + \Delta G_{tx} + \Delta G_{rx} - 2\Delta d$$

Factor	Impact
Transmit Power	3 dB
Transmitter Gain	3 dB
Receiver Gain	3 dB
Distance	6 dB

Effect of angle:

- Pointing loss (in dB) **varies quadratically** with pointing mismatch.
- So, a 3 dB change in the angle (2 times in linear scale) changes the pointing loss by 4 times.
- This can contribute sharply to the losses.
- For an antenna with a half-power bandwidth of 30 degrees:



Sensitivity to changes

Along with these, the modulation scheme also affects the link margin for a given bit error rate.

We can find a threshold value of Eb//N0 for a given BER and modulation scheme. We assume a **BER of** 10^{-5} here

BSPK & QPSK

$$P_b = Q(\sqrt{\frac{2E_b}{N_0}})$$

$$E_b/N_0 = \frac{2(Q^{-1}(P_b))^2}{2} \approx 9.6 \text{ dB}$$

8 PSK

$$P_b \approx 2Q\left(\sqrt{2\frac{E_b}{N_0}} sin(\pi/8)\right)$$

$$E_b/N_0 \approx 14 \text{ dB}$$

16 QAM

$$P_b \approx \frac{3}{8}Q(\sqrt{\frac{4}{5}\frac{E_b}{N_0}}$$

$$E_b/N_0 \approx 16 \text{ dB}$$

The link margin is calculated on the basis of this as the minimum threshold.

MATLAB Interactive Link Budget Tool

Parameter Sliders

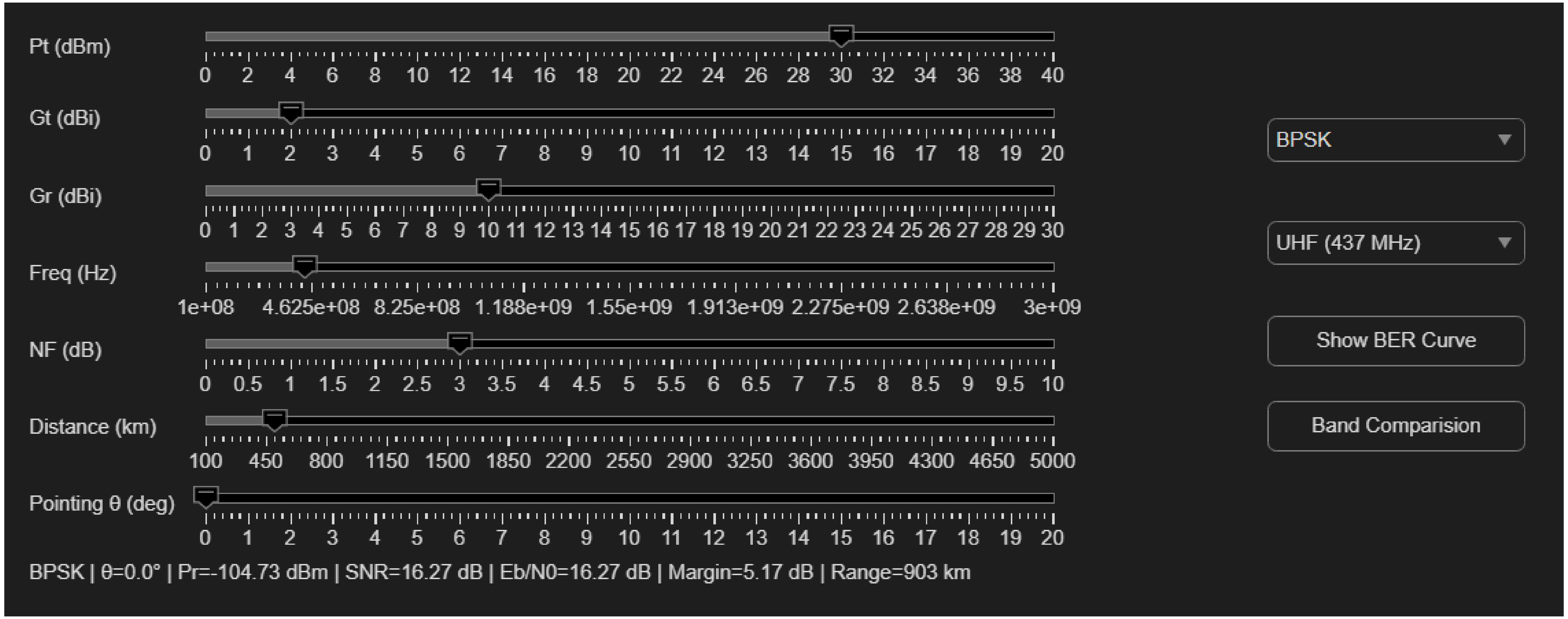
- Transmit Power Pt
- Antenna Gains Gt, Gr
- Frequency
- Distance
- Noise Figure
- Pointing Angle

Dropdown Selectors

- Modulation:
  BPSK/QPSK
  8PSK
  16QAM
- Band:
  UHF /S-Band

Real-Time Outputs

- Received Power Pr
- SNR
- Eb/N0
- Link Margin
- Maximum Range



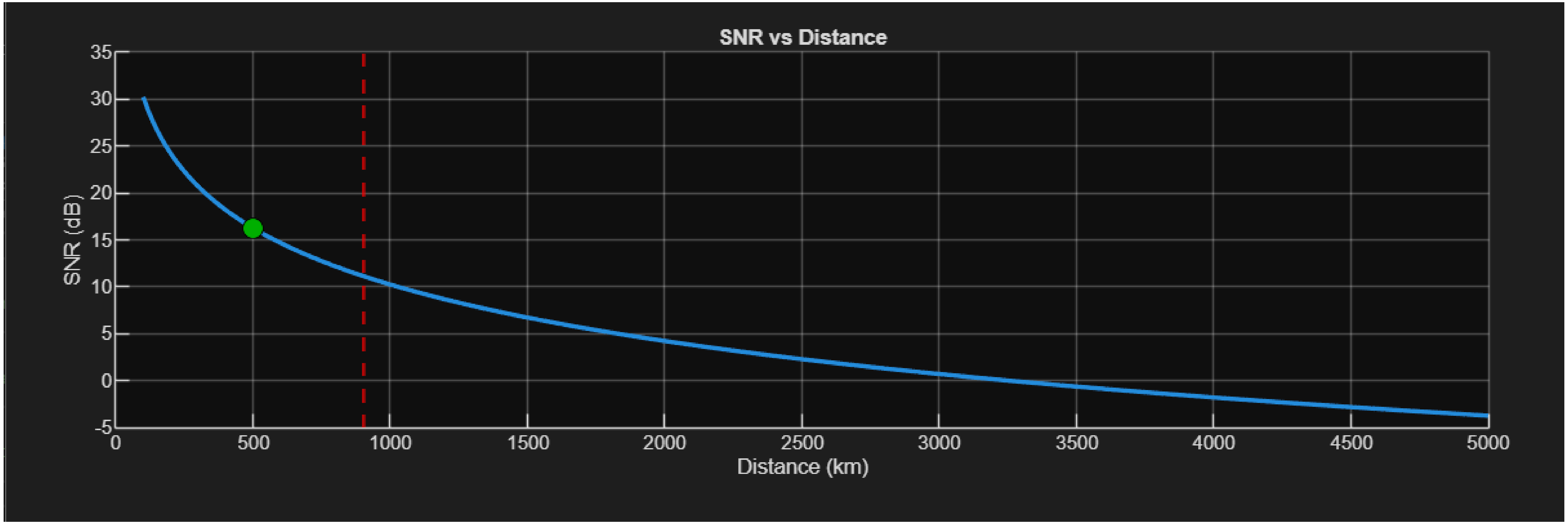
SNR vs Distance & Maximum Link Range

What is plotted

- SNR vs distance (100–5000 km)
- Current operating point (colored marker)
- Maximum range (vertical line)

Observation

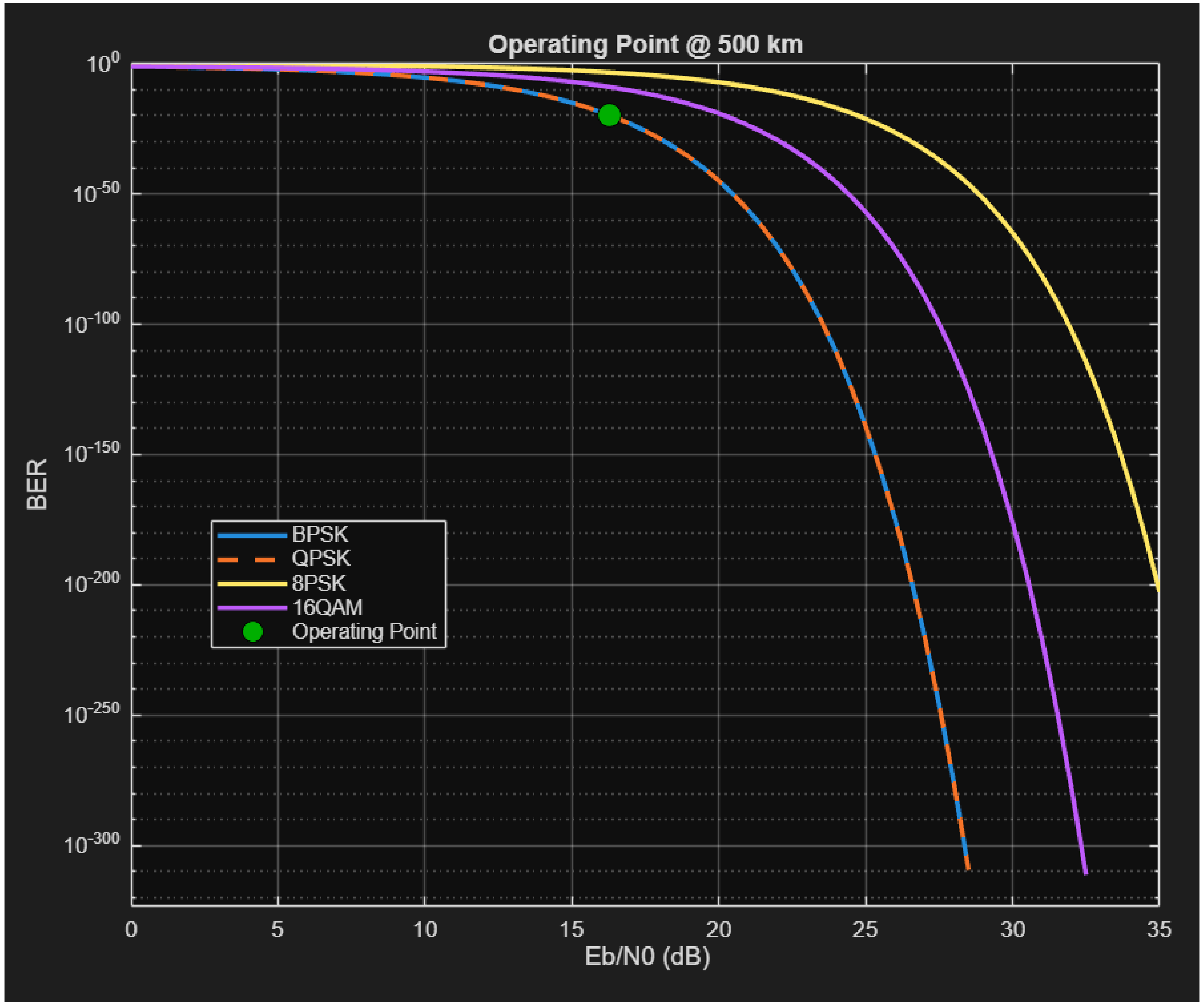
- SNR decreases with distance due to FSPL
- Link range defined where: Margin≥0



Marker color:

- Reliable - green
- Marginal - yellow
- Link failure - red

BER Performance

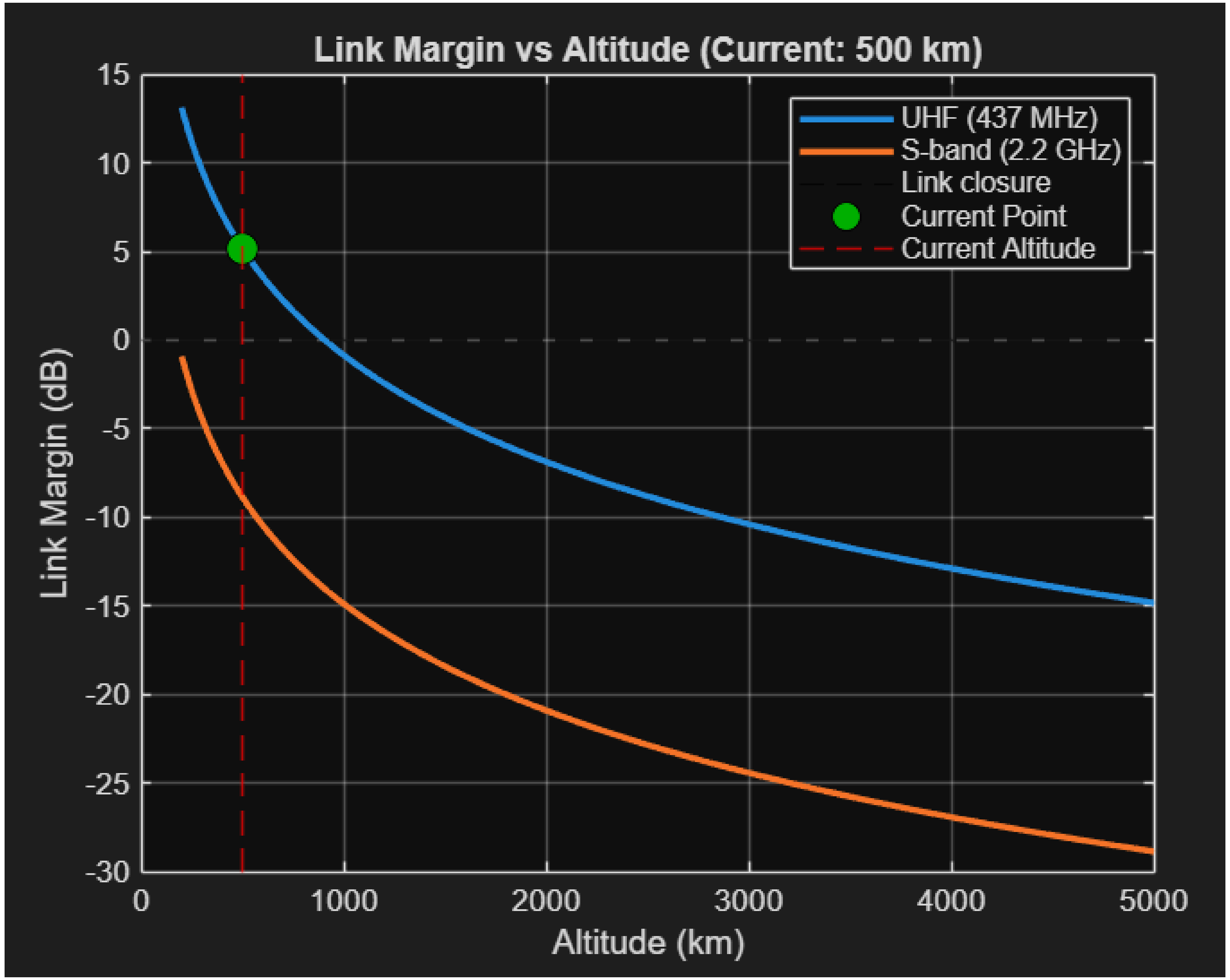


Observations

- Lower-order modulation → lower BER
- Higher-order modulation → requires higher Eb/N0

- Distance ↑ → Pr ↓ → Eb/N0↓
- Operating point shifts → BER worsens

Link Margin vs Altitude: UHF vs S-Band



Observation

1. The UHF link is viable at 500 km
2. S-band link fails at the same distance
3. Strong frequency dependence:
   Difference ≈ 14 dB at 500 km
4. S-band requires higher antenna gain to match UHF performance.

Thank You

Appendix - FSPL Difference Calculation

$$FSPL = 20\log_{10}(d) + 20\log_{10}(f) + 32.44$$

Difference due to frequency:

$$\Delta FSPL = 20\log_{10}\left(\frac{f_S}{f_{UHF}}\right)$$

Substituting values:

$$f_{UHF} = 437\,\mathrm{MHz}, \quad f_S = 2200\,\mathrm{MHz}$$

$$\Delta FSPL = 20\log_{10}\left(\frac{2200}{437}\right) = 20\log_{10}(5.03)$$

$$\Delta FSPL \approx 20 \times 0.70 \approx 14\,\mathrm{dB}$$

Appendix - Uplink noise temperature calculation

The calculation of noise temperature on the ground antenna is relatively straightforward, with Earth's temperature taken to be 290K

For a downlink scenario, 2 cases exist:

1. If the antenna on the CubeSat is directional and pointed towards Earth, its antenna temperature is also taken as 290K
2. If there is a dipole antenna present looking at both the ground and the space. The antenna temperature is taken to be about half of the temperature on Earth (150K)

$$N_{density} = k_b T$$

$$N_{density,dB} = 10\log_{10} k_b + 10\log_{10} T$$

at 290K:

$$N_{density,dB} = -228.6 + 24.62 \approx -204\text{ dB} = -174\text{ dBm}$$

Appendix - S - band